INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME

CLASS | GROUP **C / H / E / M** | INDEX NUMBER

CHEMISTRY

8873/02

27 Aug 2018

2 hour

Paper 2 Structured Questions

Candidates answer on the question paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.
Write in dark blue or black pen.
You may use pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the space provided.

Section B
Answer **1 out of 2** questions on writing paper provided..

A *Data Booklet* is provided.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [ ] at the end of each question or part question.

Paper 2	67
Paper 1	33
Overall	100
Overall Grade	

For Examiner's Use	
Section A	
1	12
2	12
3	8
4	14
5	14
Section B (Please circle)	
6 or 7	20
Significant figures and units	
Handwriting	
P2 Total	80
P1 Total	30

This document consists of **20** printed pages.

Section A

Answer all the questions.

1 Magnesium, the eighth most abundant element on Earth, has diverse applications such as in building or medicine.

(a) **(i)** Write equations for the reactions of the oxides of magnesium and phosphorus with water.

…………………………………………………………………………………………………..

…………………………………………………………………………………………...[2]

(ii) Write equations for the reactions of magnesium chloride and phosphorus chloride with water respectively. Include in your answer, the pH of the resultant solutions.

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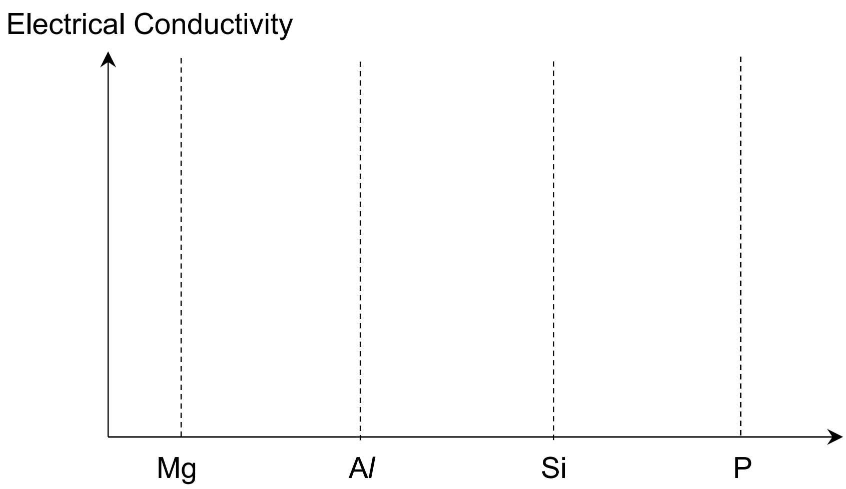
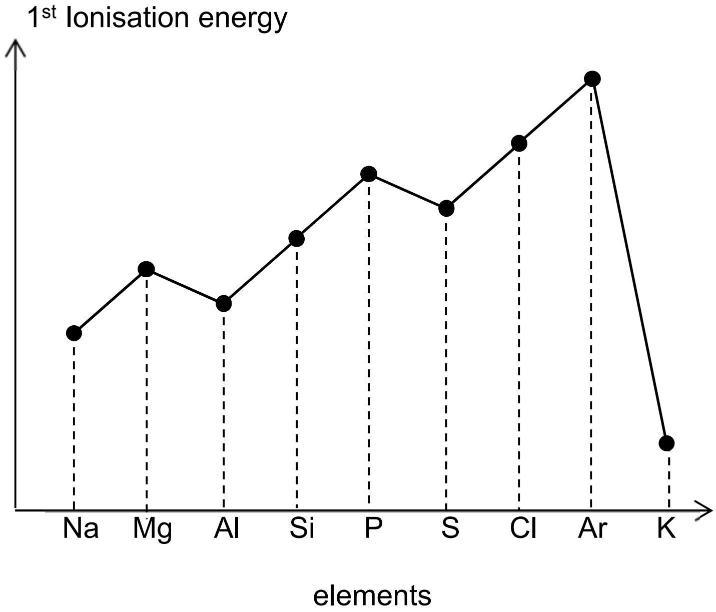
…………………………………………………………………………………………...[4]

(iii) On the axes below, sketch the electrical conductivity of the period 3 elements from magnesium to phosphorous.



[1]

(b) The following graph shows the trend of the first ionisation energies of the elements from sodium to potassium.



(i) Explain the difference between the values of their first ionisation energies for **each** of the pairs of elements listed below. You should use a different explanation for each pair.

- sodium and potassium

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……………………………………………………………………………………………..

…………………………………………………………………………………………….[2]

- magnesium and aluminium

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…………………………………………………………………………………………….[1]

(ii) When a beam of $_{20}Ca^{2+}$ particles travels through a uniform electric field which is at right angles to its direction of travel, it is deflected at an angle of +10.0°.

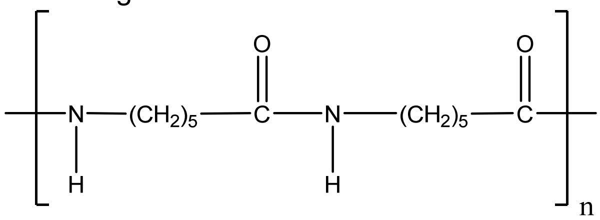
Determine the angle of deflection of a beam of $_{8}O^{2-}$ particles if it travels at the same speed through the same electric field.

[2]

[Total: 12]

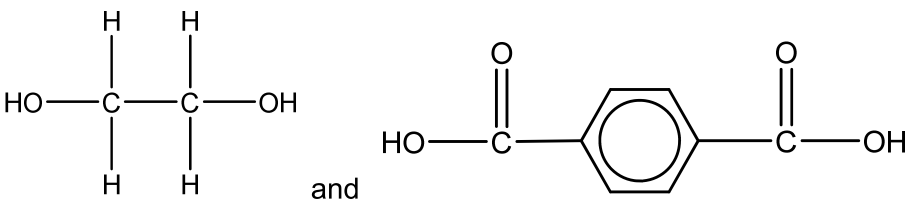
2 Synthetic polymer fibres are man-made, formed entirely by chemical synthesis. Demand for synthetic fibres has increased over the last few decades as they are durable, less expensive and more stain resistant than natural fibres. Terylene and Nylon 6 are examples of polymeric synthetic fibre.

- Nylon 6 has the following structure.



Nylon 6

- Terylene is made from ethane-1,2-diol and benzene-1,4-dicarboxylic acid:



(a) Draw the structural formula of the monomer from which Nylon 6 could be made. State the type of polymerisation involved.

Type of polymerisation: .... [2]

(b) Draw a displayed formula to show the simplest repeat unit of Terylene.

[1]

(c) Explain why nylon is prone to creasing while Terylene is wrinkle free.

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………………………………………………………………………………………………… [2]

Poly(propene) is also a polymer like the synthetic fibre; it is lightweight, durable and widely used in industries for making home articles such as container, buckets and crates.

(d) Explain why dilute sodium hydroxide will cause holes to appear in containers made from polymers such as Terylene but a poly(propene) container can be used to store sodium hydroxide.

…………………………………………………………………………………………………

………………………………………………………………………………………………… [1]

(e) State the reason why the disposal of poly(propene) containers in landfill sites is a problem.

…………………………………………………………………………………………………

………………………………………………………………………………………………… [1]

(f) The following is an extract from an article "The safety of *nanoparticles* in sunscreens: An update for general practice" published on The Royal Australian College of General Practitioners 2016, June 2016.

"Recent media coverage has raised public awareness regarding the safety of sunscreens containing zinc (ZnO) and titanium (TiO_2) *nanoparticles*. The inorganic filters ZnO and TiO_2 are able to protect against UVA and UVB. As conventional micronised particles in sunscreen, they have several unfavourable characteristics, including difficulty in application, leaving white residue on the skin. This led to the development of smaller *nanoparticles*, the use of which vastly improves aesthetic and ease of application of sunscreens while maintaining photo-protective qualities."

(i) Define *nanoparticles*.

…………………………………………………………………………………………. [1]

(ii) Explain in your own words why the surface-to-volume ratios makes nanoscale sunscreen attractive.

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………………………………………………………………………………………………

…………………………………………………………………………………………. [2]

(iii) Explain why nanoparticles of TiO_2 in sunscreen could present a risk to human health.

…………………………………………………………………………………………. [1]

(iv) Sunscreen containing nanoparticles should be tested further.
Suggest one reason why some companies that make sunscreens might not want to do more tests.

………………………………………………………………………………………………

…………………………………………………………………………………………. [1]

[Total: 12]

3 Citric acid, $C_6H_8O_7$, is a weak tribasic organic acid. It is used as an additive and preservative in various food and drinks to enhance flavour and taste. It provides a fruity tartness to complement fruit flavours and it is also used to mask the unpleasant taste of pharmaceutical products.

However, a high content of citric acid in foods and drinks could damage your teeth. Chemical erosion of the teeth occurs either by the hydrogen ions derived from citric acid or by the citric acid anions which can bind or complex calcium. The acceptable limit for the percentage composition by mass of citric acid in food is 2.5% as this is around the percentage composition commonly found in natural fruits.

A student wanted to find the out the mass of citric acid in a mint candy by doing a titration with sodium hydroxide solution. The equation for the titration is:

$$C_6H_8O_7 + 3NaOH \rightarrow C_6H_5O_7Na_3 + 3H_2O$$

He weighed out 5.00 g of the mint candy and crushed the candy into powder. He then dissolved the powder in 100 cm^3 of distilled water, stirred the mixture and filtered it forming solution **A**.

(a) The student then prepared a standard solution of sodium hydroxide by dissolving 0.104 g of sodium hydroxide pellets in 25.0 cm^3 of distilled water. He then made the volume up to 250 cm^3 forming solution **B**.

Calculate the concentration of sodium hydroxide in solution **B**.

[2]

(b) The student then found that 20.0 cm^3 of solution **A** required 10.40 cm^3 of solution **B** for a complete reaction.

(i) Suggest a suitable indicator for the titration.

.... [1]

(ii) Calculate the mass of citric acid in the 20.0 cm^3 of solution **A**.

[2]

(iii) Given that the average weight of one piece of mint candy is 0.50 g, calculate the mass of citric acid in one piece of mint candy.

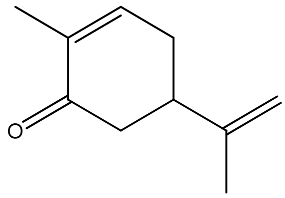
[2]

(c) From your answer in **(b) (iii)**, determine whether the percentage composition by mass of citric acid in the mint candy is above or below the acceptable limit.

[1]

[Total: 8]

4. Carvone is a natural oil produced by the plant *Mentha spicata.* It has the flavour of spearmint and it has important uses in food flavouring.



Carvone

(a) **(i)** Identify the functional groups present in Carvone.

……………………………………………………………………………………….[2]

(ii) Draw structures for the organic products of each compound with the following reagents. State the type of reaction in each case.

Reagent and conditions	**Type of reaction**	**Organic product**
Br_2 liquid, room temperature		
H_2 with Ni catalyst, high temperature and high pressure		
$LiAlH_4$ in dry ether, room temperature		

[7]

(b) Suggest why Carvone is highly soluble in hexane.

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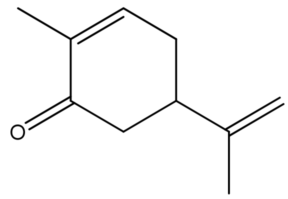
……………………………………………………………………………………………… [2]

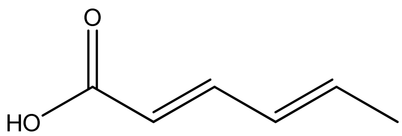
(c) On the above structure of Carvone, circle one sp^2 hybridised carbon atom.



Carvone

[1]

(d) Sorbic acid, or 2,4-hexadienoic acid, is a natural organic compound used as a food preservative.



Sorbic acid

Describe a chemical test that will help you distinguish Carvone from Sorbic acid, including all observations that would be made.

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……………………………………………………………………………………………… [2]

[Total: 14]

5 **(a)** Benzene undergoes complete combustion with oxygen as shown by the equation below.

$$C_6H_6(l) + O_2(g) \rightarrow CO_2(g) + H_2O(l)$$

(i) Explain what is meant by *standard enthalpy change of combustion.*

………………………………………………………………………………………………

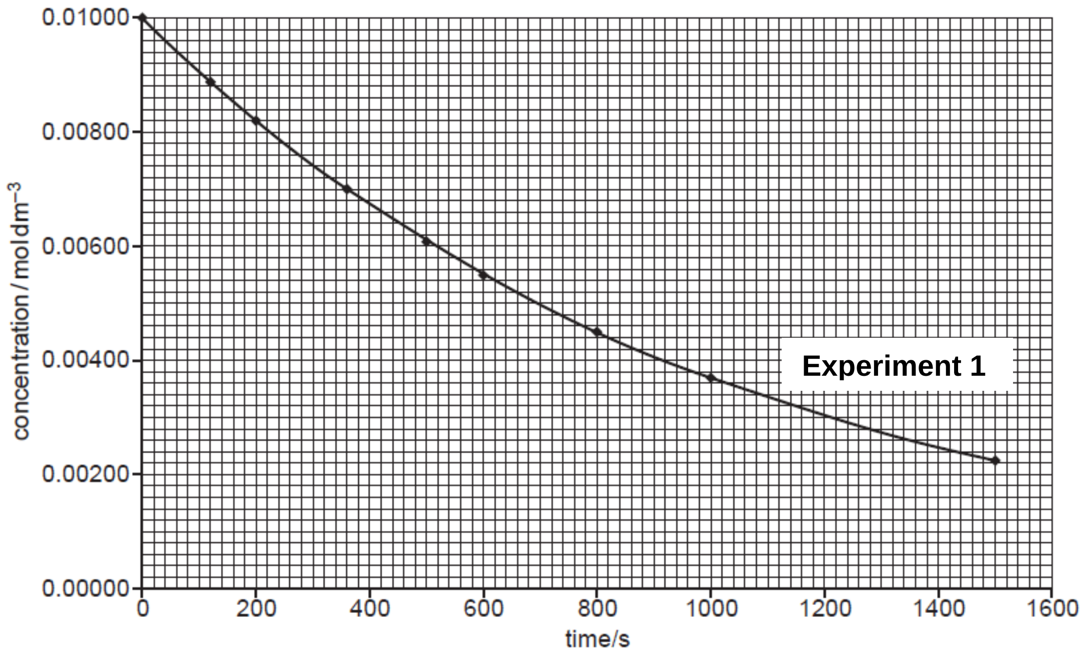
………………………………………………………………………………………...[1]

(ii) 1.17 g of benzene was burnt to heat up 250 g of water in a beaker. The initial temperature of the water was 27.6°C. Given that the standard enthalpy change of combustion of benzene is –3267 kJ mol^{-1} is and there is 20% heat loss to the surrounding, calculate the highest temperature reached by the water.

[3]

(b) 2–iodo–2–methylpropane undergoes a substitution reaction with hot aqueous sodium hydroxide. Two separate experiments were carried out to study the kinetics of this reaction.

In Experiment 1, sodium hydroxide was used in large excess and the concentration of 2–iodo–2–methylpropane was measured against time. A graph of concentration of 2–iodo–2–methylpropane against time was plotted and shown below.



(i) Use the half–life method to deduce the order of reaction with respect to 2–iodo–2–methylpropane. Show all your working clearly.

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…………………………………………………………………………………………………..

……………………………………………………………………………………………….[2]

In Experiment 2, 2–iodo–2–methylpropane was used in large excess and the concentration of sodium hydroxide was measured against time. The following results were obtained.

Time / s	**[NaOH] / mol dm^{-3}**
0	0.0100
100	0.0082
200	0.0064
380	0.0032
500	0.0010

(ii) Using the data provided above, plot a graph of [NaOH] vs time for Experiment 2 in the graph above.

[1]

(iii) Using the graph plotted in **(b)(ii)**, determine the order of reaction with respect to sodium hydroxide. Explain your answer.

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…………………………………………………………………………………………………[2]

(iv) Hence, write an overall rate equation for the substitution reaction.

…………………………………………………………………………………………………[1]

(c) Colourless N_2O_4 and brown NO_2 are two oxides of nitrogen that can co-exist in equilibrium:

$$N_2O_4\,(g) \rightleftharpoons 2\,NO_2\,(g) \qquad \Delta H = +\,58\ \text{kJ mol}^{-1}$$

Predict and explain how the colour intensity of the mixture changes, when a reaction chamber containing an equilibrium mixture of N_2O_4 and NO_2 is:

(i) immersed in boiling water.

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………………………………………………………………………………………………….

…………………………………………………………………………………………………[2]

(ii) subjected to a sudden compression in volume.

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…………………………………………………………………………………………………[2]

[Total: 14]

Section B

Answer **one** question from this section, in the spaces provided.

6 **(a)** **(i)** Describe the structure of graphite. Explain how this structure gives graphite the properties of soft and high electrical conductivity.

You may include labelled diagrams in your answer.

....

....

....

....

....

....[3]

(ii) Another allotrope of carbon is diamond. State one physical property diamond has in common with graphite and relate the property to the bonding present in their structures.

....

....

....[2]

(b) **(i)** Write an equation, including state symbols, for the reaction with enthalpy change equal to the standard enthalpy of formation for ethane, C_2H_6(g).

....[1]

(ii) Suggest in terms of bonding why ethane is a gas at room temperature.

....

....[1]

(iii) Ethane undergoes a reaction with chlorine. Name the type of reaction that occurs and the conditions used.

....

....[2]

(c) The enthalpy change for the following reaction is –2889 kJ mol^{-1}.

$$C_2H_6(g) + 7F_2(g) \longrightarrow 2CF_4(g) + 6HF(g)$$

Use this value and the standard enthalpies of formation given to calculate the standard enthalpy of formation of $C_2H_6(g)$.

Substance	$CF_4(g)$	$HF(g)$
ΔH_f^o / kJ mol^{-1}	–680	–269

[2]

(d) Methane reacts violently with fluorine according to the following equation.

$$CH_4(g) + 4F_2(g) \longrightarrow CF_4(g) + 4HF(g) \qquad \Delta H = –1904 \text{ kJ mol}^{-1}$$

(i) Using relevant bond energies in the Data Booklet, calculate the bond energy of F–F bond.

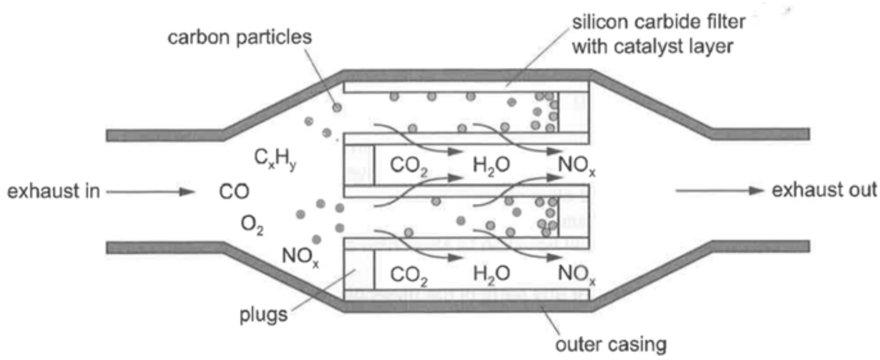
[2]

(ii) A student suggested that actual bond energy of F-F differs from theoretically calculated value in **(i)**. Explain if you agree or disagree with the student.

....

....[1]

(e) Catalyst are used in car exhaust of modern cars to speed up reaction between polluting gases such as carbon monoxide and dinitrogen oxide, before they reach the end of the exhaust pipe.



(i) State the type of catalyst involved in this reaction.

.... [1]

(ii) Sketch a Boltzmann distribution curve for the reactants and use it to explain how the use of catalyst speeds up the reaction.

Explanation

....

....

....

....

....[4]

(iii) Suggest why car manufacturers are required to fit catalytic converters to car exhaust systems.

....

.... [1]

[Total: 20]

7 **(a)** Phosphorus reacts with fluorine to form PF_3 and PF_5.

(i) Draw the dot-and-cross diagram for one molecule of PF_5. Show only the outer shell electrons.

[1]

(ii) Explain why the P–F bond is polar.

…………………………………………………………………………………………….. [1]

(iii) Predict whether or not a molecule of PF_5 is polar. Explain your answer.

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…………………………………………………………………………………………….. [1]

(iv) Nitrogen reacts with fluorine to form NF_3 only.

Explain why PF_5 exists whereas NF_5 does not exist.

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…………………………………………………………………………………………….. [1]

(b) The reaction between sulfur dioxide and oxygen is a reversible reaction and can be described as being in dynamic equilibrium.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

A 2-dm^3 reaction vessel containing 0.80 mol of SO_2, 0.30 mol of O_2 and 1.40 mol of SO_3 is allowed to reach equilibrium at a constant temperature of 1000 K. It was determined that 0.42 mol of SO_2 is present at equilibrium.

(i) State the meaning of the term *dynamic equilibrium*.

……………………………………………………………………………………………

…………………………………………………………………………………………….. [1]

(ii) Use the information given to complete Table 6.1.

Table 6.1

	SO_2	O_2	SO_3
Initial amount/ mol			
Equilibrium amount/ mol			

[2]

(iii) Write an expression for the equilibrium constant K_c.

[1]

(iv) Calculate the value of K_c. State the units of K_c.

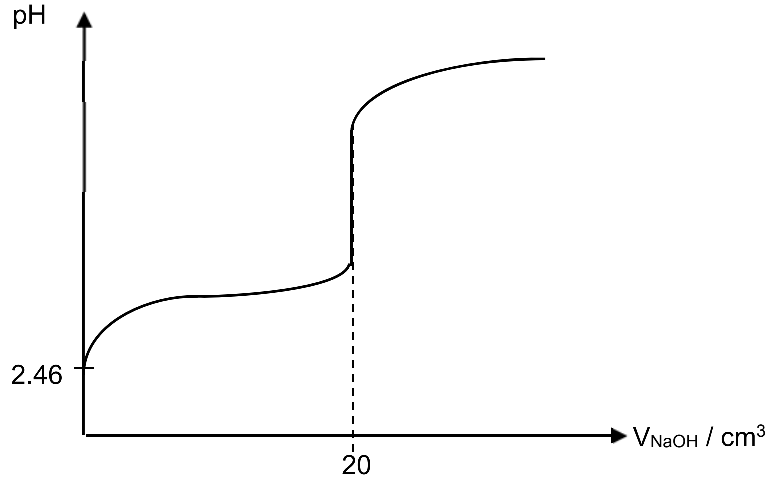
[2]

(c) (i) State the difference between a weak acid and a strong acid.

………………………………………………………………………………………………

……………………………………………………………………………………….. [1]

20.0 cm^3 of 0.01 mol dm^{-3} propanoic acid, CH_3CH_2COOH was titrated against 0.01 mol dm^{-3} aqueous sodium hydroxide.



The initial pH of propanoic acid is 2.46.

(ii) Calculate the initial concentration of hydrogen ions before the titration.

[1]

During the addition of the first 20 cm^3 of sodium hydroxide the mixture is behaving as a buffer.

(iii) Explain how the shape of the graph show this.

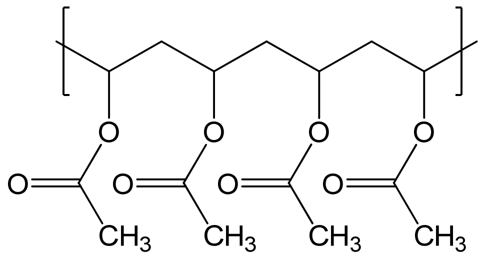
…………………………………………………………………………………………………

…………………………………………………………………………………………….. [1]

(iv) Identify the two species present in the mixture that constitutes to it behaving as a buffer.

…………………………………………………………………………………………….. [1]

(d) Polyvinyl acetate, PVAc, is a useful adhesive for gluing together articles made from wood, paper or cardboard. The diagram shows a section (not a repeat unit) of the PVAc.



PVAc

(i) What type of polymerisation made this polymer?

…………………………………………………………………………………………….. [1]

(ii) PVAc is made from monomer **X**. Draw the structure of the **X**.

[1]

(iii) PVAc can be converted to polyvinyl alcohol, PVA.

Draw a section of PVA containing at least 2 repeat units and suggest the reagents and conditions needed to make PVA from PVAc in the laboratory.

Section of PVA:

Reagents and conditions: ……………………………………………………………..
[2]

(e) The diagrams below are molecular representations of two types of plastics – thermoplastics and thermosets.

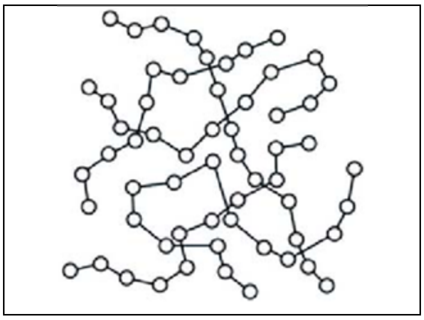


Diagram **A**

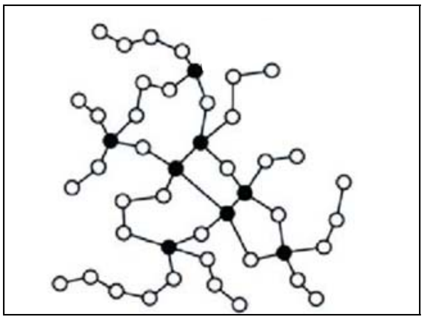


Diagram **B**

(i) State and explain which diagram could be the representation for thermosets.

…………………………………………………………………………………………

………………………………………………………………………………………….. [1]

(ii) State a difference in physical properties between thermoplastics and thermosets.

…………………………………………………………………………………………

………………………………………………………………………………………….. [1]

[Total: 20]

End of Paper